

Surgical Suture Instrument Detection System

ระบบตรวจจับอุปกรณ์การเย็บแผลด้วยปัญญาประดิษฐ์

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PROJECT SUMMARY

This project presents the development of an AI-based object detection system using the YOLOv11 Object Detection model to detect surgical instruments and related objects from video data. The system is designed to identify five object classes: needle, finger, needle holder, wound, and forceps. The development workflow consists of video preprocessing, frame extraction, data annotation, model training, and performance evaluation. After training, the model is capable of accurately identifying and classifying objects within video frames. The results demonstrate the effectiveness of the YOLOv11 model for object detection tasks and highlight its potential as a reliable foundation for future computer vision research and intelligent object detection applications.

PROBLEM STATEMENT

Object detection from video data is a challenging task because objects often vary in size, orientation, lighting conditions, and movement, while some objects may also be partially occluded. These factors can reduce the accuracy and reliability of object detection systems, especially when multiple object classes appear simultaneously within a single frame. Therefore, an efficient and robust object detection model is required to accurately identify and classify objects in real time. This project addresses these challenges by utilizing the YOLOv11 Object Detection model to detect five object classes from video frames and evaluate its detection performance using standard object detection metrics.

OBJECTIVES

- To develop an AI-based object detection system using the YOLOv11 model.
- To detect five object classes: needle, finger, needle holder, wound, and forceps from video frames.
- To evaluate the performance of the object detection model using standard evaluation metrics, including Precision, Recall, and mAP@50.

IMPACT

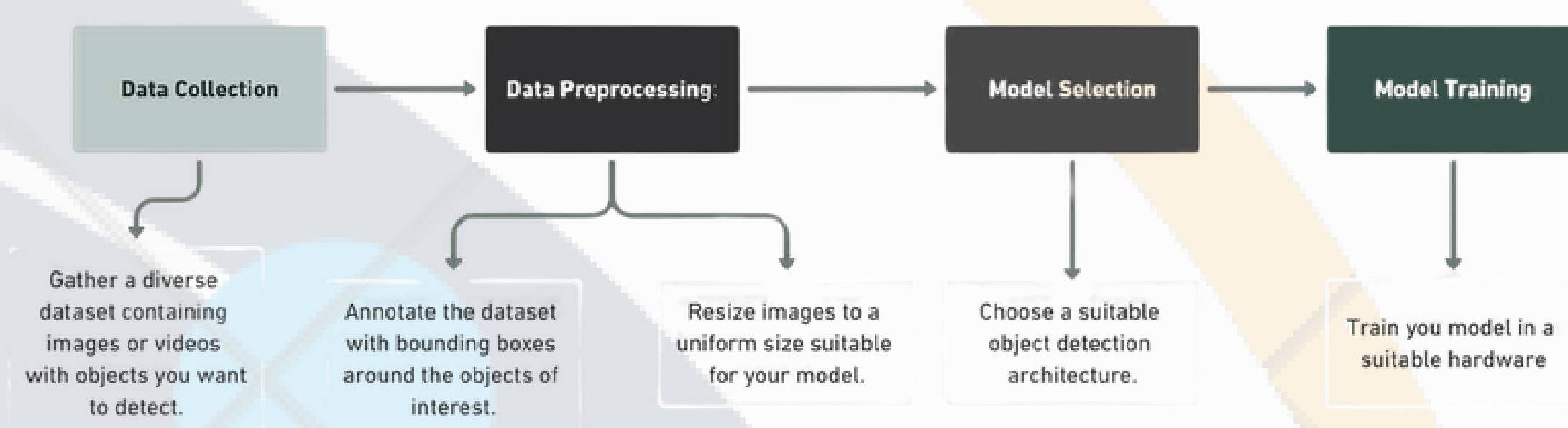
Demonstrates the effectiveness of the YOLOv11 model for real-time object detection from video data.

Provides a structured workflow for developing AI-based object detection systems including data preprocessing, annotation, model training, and evaluation.

Serves as a foundation for developing more advanced intelligent object detection systems.

Supports future research in computer vision and deep learning for object detection applications.

METHODOLOGY



RESULTS

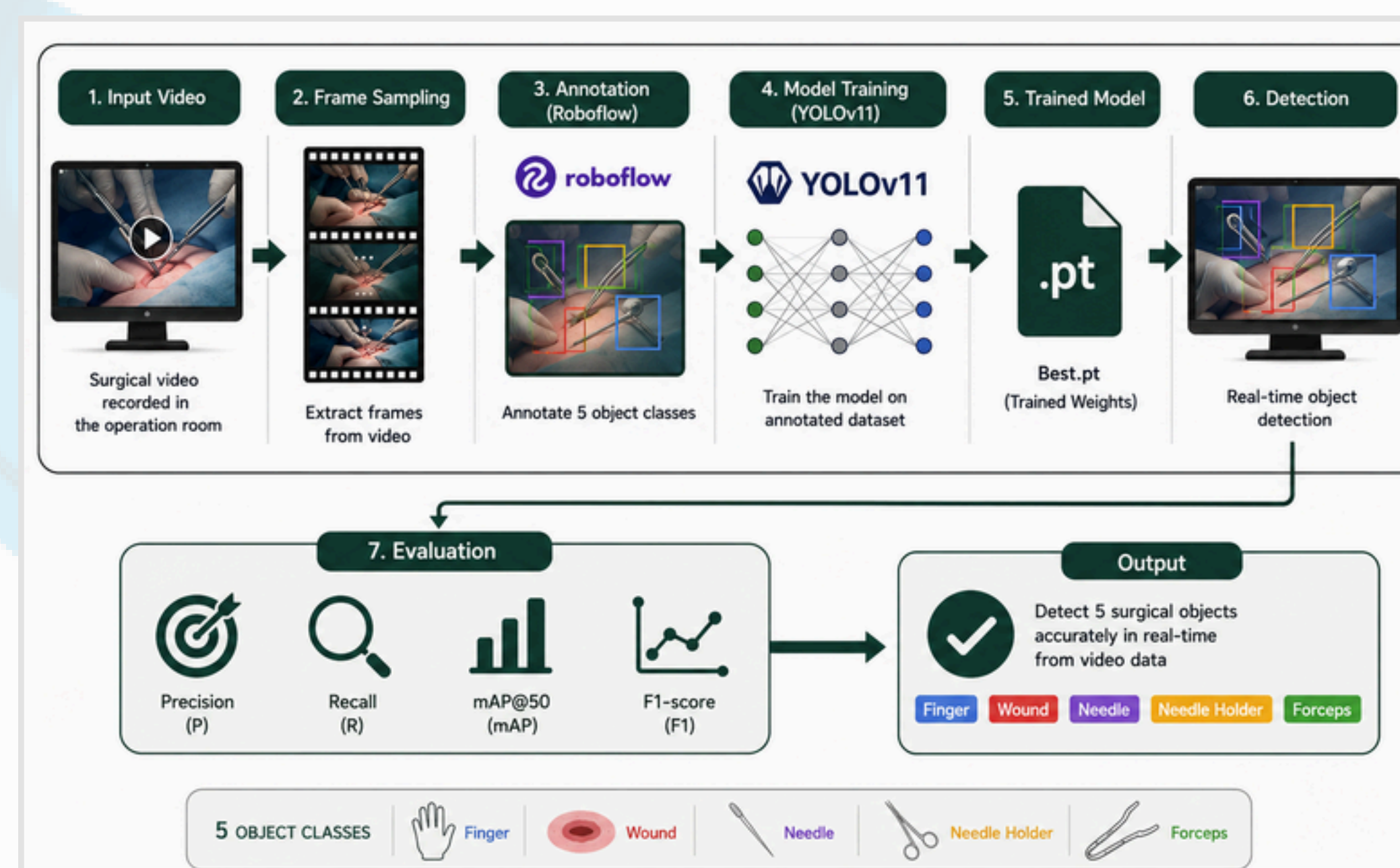
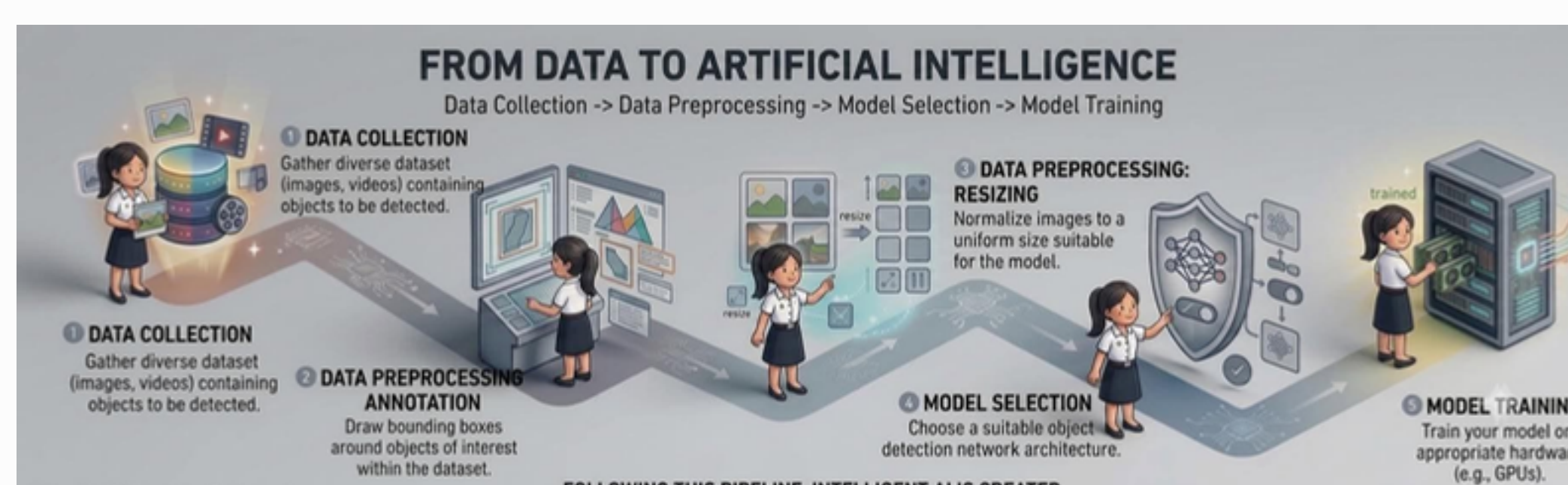
The suture tool detection system developed using the YOLOv11 model is capable of detecting all 5 target object categories—fingers, wounds, needles, needle holders, and forceps—from images extracted from suturing videos. This demonstrates that the system can simultaneously detect multiple types of objects, and the detection results show that the model possesses a high capability for precise object localization, holding strong potential for future applications in surgical assistance and medical equipment detection systems.

CONCLUSION

This project demonstrates the development of an AI-based surgical suture instrument detection system using the YOLOv11 object detection model. The complete workflow, including data preparation, annotation, model training, and performance evaluation, provides an effective approach for detecting and classifying five types of surgical instruments and related objects. The results indicate that the developed model has strong potential to serve as a foundation for future intelligent surgical assistance systems and other computer vision applications.

ENGINEERING DESIGN / SYSTEM ARCHITECTURE

The system starts by collecting suture video footage, then performing frame sampling and data refinement before performing annotations on Roboflow to create a dataset for training the YOLOv11 model. After training, the model can detect all five types of devices in real-time, and the trained model is evaluated using Precision, Recall, F1-score, and mAP@50 values.



INTEGRATION COURSES

COE64-367 Special Topics in Data Analytics II
COE64-342 Deep Learning
COE64-307 Artificial Neural Networks
COE64-361 Convolutional Neural Networks
COE64-371 Front End Programming

QR CODE

